

The Accidental Computer: Polynomial Commitments from Data Availability

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Abstract

In this paper, we present two simple variations of a data availability scheme that allow it to function as a multilinear polynomial commitment scheme over the data in a block. The first variation enables commitments over all of the block’s data with zero prover overhead: the data availability construction simply serves both purposes. The second variation allows commitments over subsets of data with nonzero but still concretely small proving costs, since most work is already done during data encoding. Since, in many succinct proof systems, opening a polynomial commitment contributes significantly to prover costs, our construction enables the prover to reuse work already done by the data availability scheme, reducing—or wholly removing—work associated with the polynomial commitment scheme.

1 Introduction

Blockchains, as their name suggests, are composed of an ordered list of blocks. These blocks may include the data associated with financial transactions, payments, or even social media posts and interactions. An important part of this construction is that, as new blocks are added to this list, these new blocks must satisfy some computational rules (or predicates) which ensure that the data in the blocks is valid. For example, one (of potentially many) rules to be enforced may be that no transaction in the block may transfer more money out of an account than the account has available; *i.e.*, that account balances may not be negative. The list of such predicates is often called the *state transition function* of a blockchain: the predicates ensure that the latest state of the blockchain, once it has been modified by the transactions of a block, is still a valid end state.

Designs. In early blockchain designs, verifying the state transition function required downloading all transaction data for each block and verifying the predicates for all downloaded blocks. As blockchains scale to support more concurrent users, and therefore bigger blocks